

ASSET DEPENDENCY MATRIX (ADM)

PART I — DEPENDENCY GOVERNANCE

Chapter 1 — Purpose

Purpose

The Asset Dependency Matrix (ADM) documents every production relationship between approved assets within Explorer Studio.

While the Master Asset Registry records **what assets exist**, the Asset Dependency Matrix records **how those assets depend upon one another**.

Its purpose is to preserve production integrity by ensuring that changes to one asset can be evaluated against every related asset before implementation.

The Dependency Matrix therefore acts as the connective infrastructure linking the entire Master Asset Library into one coherent production ecosystem.

Objectives

The Asset Dependency Matrix exists to:

- document asset relationships,
- prevent hidden production dependencies,
- support impact analysis,
- simplify maintenance,
- improve reuse,
- reduce production risk,
- support automation,
- preserve production knowledge.

Guiding Principle

No asset exists in isolation.

Every approved asset participates in one or more production relationships.

Understanding those relationships is essential for sustainable long-term production.

Chapter 2 — Dependency Philosophy

Explorer Studio distinguishes between **assets** and **relationships**.

Assets create capability.

Relationships create production.

Therefore, relationships receive the same engineering discipline as assets themselves.

Dependency Rules

Every approved dependency shall be:

- documented,
- traceable,
- reviewable,
- maintainable,
- version controlled.

Undocumented dependencies are considered production defects.

PART II — DEPENDENCY ARCHITECTURE

Chapter 3 — Dependency Categories

Explorer Studio recognizes the following permanent dependency categories.

Structural Dependency

Defines physical composition.

Example

Grand Archive
↓
Reading Hall
↓
Master Reading Table

Functional Dependency

Defines operational behavior.

Example

Lantern
↓
Lighting Preset
↓
Shadow Profile

Visual Dependency

Defines aesthetic consistency.

Example

Guild Cloak
↓

Guild Color Palette



Brand Identity

Material Dependency

Defines shared physical materials.

Example

Oak Wood



Reading Table



Bookshelf



Bridge

Animation Dependency

Defines animation reuse.

Example

Walking



Explorer

Keeper

Archivist

Behavioral Dependency

Defines simulation relationships.

Example

Wind



Trees

Grass

Clothing

Lantern Flame

Water

Environmental Dependency

Defines ecosystem relationships.

Example

River



Fish

Reeds

Dragonflies

Birds

Story Dependency

Defines narrative relationships.

Example

Legendary Journal



Missing Pages



Living Record

Production Dependency

Defines workflow relationships.

Example

Concept Art



Model



Texture



Rig



Animation

Franchise Dependency

Defines recurring Explorer Studio identity.

Example

Opening Identity



Guild Logo



Archive Theme



Living Record Transition

Chapter 4 — Dependency Direction

Dependencies always have direction.

Parent Asset



Dependent Asset

The parent supplies capability.
The dependent consumes capability.

Chapter 5 — Dependency Strength

Every dependency receives one strength classification.

★★★★★
Critical
Cannot function independently.

★★★★☆
Strong
Major influence.

★★★☆☆
Moderate
Partial dependency.

★★☆☆☆
Weak
Occasional relationship.

★☆☆☆☆
Informational
Reference only.

PART III — MASTER DEPENDENCY MATRICES

Environment Matrix

Parent	Child	Type	Strength
Grand Archive	Reading Hall	Structural	★★★★★
Reading Hall	Reading Table	Structural	★★★★★
Reading Hall	Lantern System	Functional	★★★★☆
Hidden Valley	Terrace Agriculture	Environmental	★★★★★
Hidden Valley	Water Distribution	Functional	★★★★★
Mountain System	Mist System	Environmental	★★★★★

Character Matrix

Parent	Child	Type
Explorer Apprentice	Travel Costume	Structural
Explorer Apprentice	Walking Animation	Animation
Explorer Apprentice	Explorer Journal	Functional
Elder Keeper	Keeper Staff	Functional
Guild Master	Guild Robes	Visual

Prop Matrix

Parent	Child	Type
Legendary Journal	Missing Pages	Story
Legendary Journal	Journal Wrap	Structural
Explorer Backpack	Rope Kit	Structural
Explorer Backpack	Water Flask	Structural

Material Matrix

Material	Assets Using
Oak Wood	Table, Shelf, Bridge
Stone	Archive, Gateway, Assembly House
Leather	Journal, Backpack, Boots
Linen	Cloaks, Journal Wrap
Iron	Hinges, Lantern, Tools

Animation Matrix

Animation	Assets
Walk	Explorer, Keeper, Archivist
Sit	Explorer, Guild Master
Observe	Explorer, Keeper
Write	Explorer, Archivist

Behavior Matrix

Behavior	Dependent Assets
Wind	Trees, Grass, Cloaks, Smoke
Rain	Stone, Soil, Roofs, Streams
Fog	Valley, Forest, River
Fire	Lantern, Fireplace

Story Matrix

Story Asset	Depends On
Missing Pages	Legendary Journal

Silver Feather Living Record
Living Record Grand Archive

Franchise Matrix

Franchise Asset	Depends On
Opening Identity	Grand Archive
Ending Identity	Living Record
Page Transition	Journal
Guild Logo	Brand Bible

PART IV — IMPACT ANALYSIS

Chapter 6 — Change Impact

Every proposed change shall trigger an impact review.

Example:

Oak Wood Material Updated



Reading Table



Archive Shelf



Bridge Kit



Assembly House Furniture



Five Productions

The system immediately identifies all affected assets.

Chapter 7 — Dependency Validation

Before approval, validate:

- Parent exists.
- Child exists.
- Relationship documented.
- No circular dependency.
- Version compatibility maintained.

PART V — FUTURE AUTOMATION

The Dependency Matrix enables future automation such as:

- Automatic impact analysis.
- Duplicate detection.
- Broken reference detection.
- Missing dependency alerts.
- Asset recommendation.
- Reuse suggestions.
- Production planning.
- AI-assisted dependency mapping.

FINAL PRINCIPLE

An asset gives the studio capability.

A dependency gives the studio coherence.

Together they create a production ecosystem.

Chief Systems Architect Review

I would **lock the Asset Dependency Matrix** as a permanent Explorer Studio infrastructure document.

At this point, your studio has a complete production data backbone:

1. **Master Asset Library (MAL)** — standards and governance.
2. **Master Asset Registry (MAR)** — inventory of approved assets.
3. **Master Asset Specifications (MAS)** — engineering record for each asset.
4. **Asset Dependency Matrix (ADM)** — relationships and impact analysis.